

JKBOSE 2026

CLASS 11TH

BIOLOGICAL CLASSIFICATION

LECTURE-05

BOTANY

ARSHID BHAT SIR



Topics To Covered

1

Kingdom Protista

INTRODUCTION

- Term protista — Coined by Ernst Haeckel.
- Kingdom protista includes — all "unicellular eukaryotes".
- Habitat: primarily aquatic.
- Kingdom protista includes →
 - ① chrysophytes ✓
 - ② Dinoflagellates ✓
 - ③ Euglenoids ✓
 - ④ Slime moulds. ✓
 - ⑤ protozoans ✓

• Boundaries of Kingdom protista — are not well defined.

Reason: members of Kingdom protista share similarities with the members of other groups.

Example:

Chrysophytes ✓

Dinoflagellates ✗

Euglenoids ✓

Slime moulds. ✓

Protozoans ✓

Photosynthetic (like plants)

Saprophytic (like fungi)

Holozoic like animals (parasitic)

≡ An organism - may be a plant to one biologist but may be a photosynthetic protista to another biologist

≡ Members of Kingdom protista possess — well defined nucleus.
and
membrane bound organelles.

≡ Reproduction: Both $\left\{ \begin{array}{l} \text{Sexual.} \\ \text{Asexual.} \end{array} \right.$

≡ Flagella: $\left\{ \begin{array}{l} +nt \text{ in some.} \\ -nt \text{ in some.} \end{array} \right.$

≡ Cilia: $\left\{ \begin{array}{l} +nt \text{ in some} \\ -nt \text{ in some} \end{array} \right.$

CHRYSTOPHYTES



≡ It includes — Diatoms and desmids

↓
very little knowledge is known about desmids.

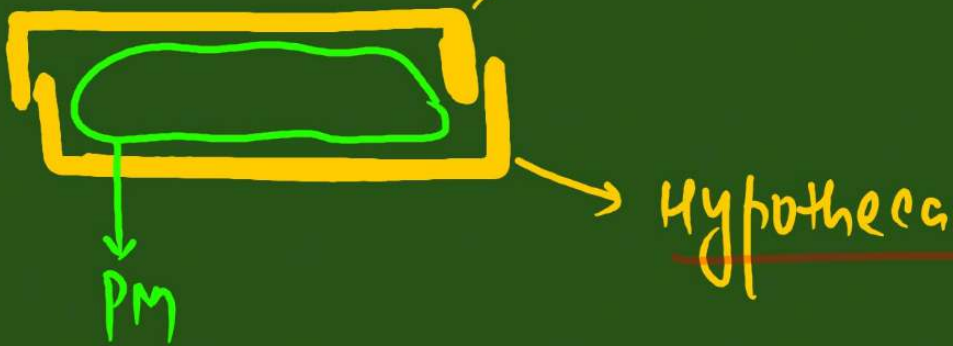
≡ Habitat:
Fresh H_2O (↓↓ Salt Conc.)
Marine H_2O (↑↑ Salt Conc.)

≡ Flagella: Absent (non-motile)

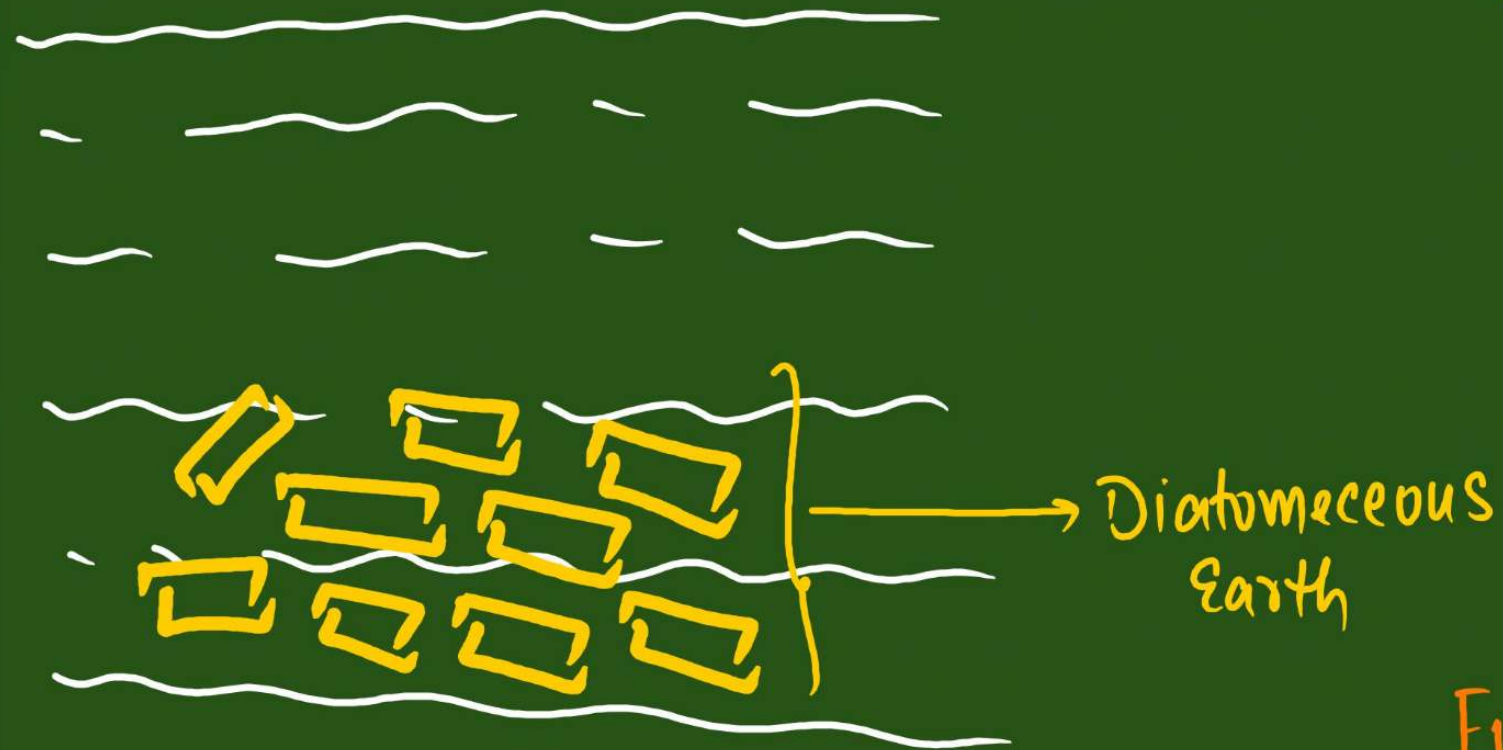
• They float passively on H_2O surface due to water currents.

• These are photosynthetic micro-organisms (plankton) — "Phytoplankton"

Cell wall:

- present.
 - Composed of Cellulose ✓ + Silica * ✓
- 
- Cell wall — possess two thin overlapping shells which fit together like a Soap box.
 - Because of +nse of silica in cell — the Cell wall of diatoms becomes indestructible.

(*) Over billions of years, when diatoms die they accumulate at bottom of water body and forms mass of cell wall kld as diatomaceous earth or Diatomite.



Functions of Diatomite:

- ① Diatomite - being gritty, it is used in polishing of metals.
- ② Diatomite - being porous in nature, it is used in filtration of oils and syrups.

Diatoms - chief producers in oceans.

Photosynthetic pigments - chl a, chl c, xanthophyll.

DINOFLAGELLATES

- Habitat: Marine H_2O ($\uparrow\uparrow$ Salt conc.)

- Cell wall: present.

• Composed of Cellulose + Pectin

Outer surface of cell wall — is composed of stiff Cellulosic plates.

- Flagella: +nt.

• 2 in no.

longitudinal flagellum
transverse flagellum.

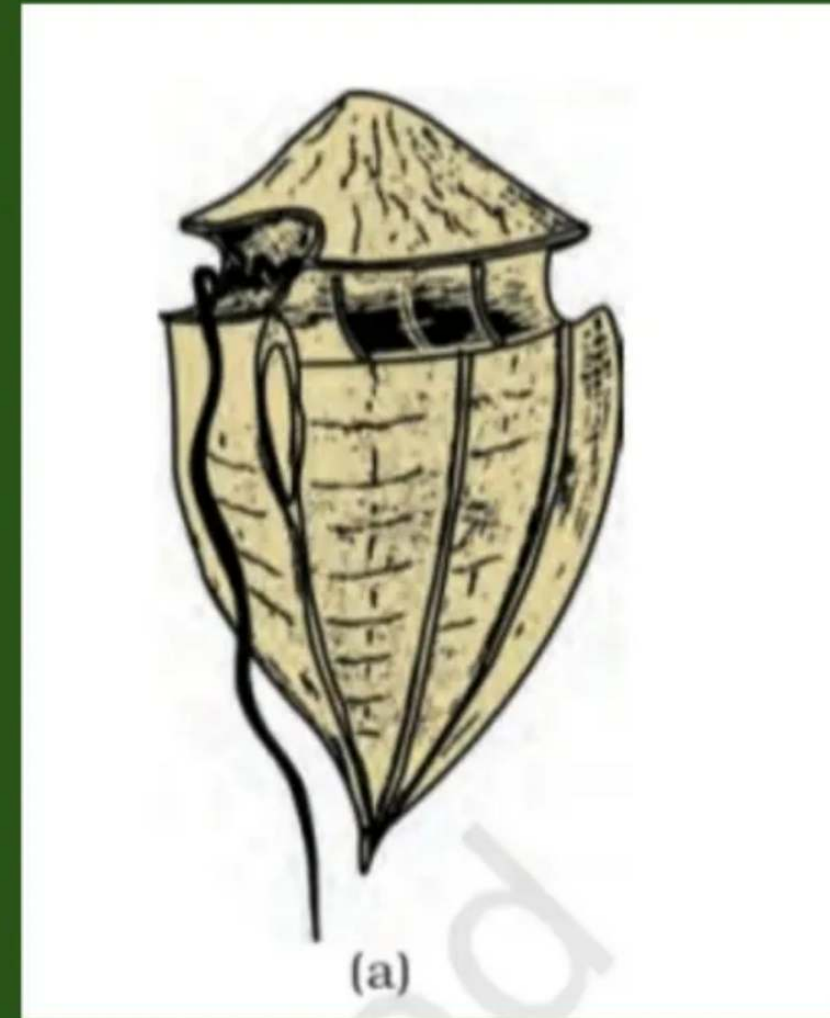
longitudinal flagellum

longitudinal groove

transverse flagellum

transverse groove

"Dinoflagellate"



≡ Dinoflagellates — may appear — Red, yellow, green, Blue etc
depending on main pigment +nt
in them.

Ex. Red dinoflagellates — possess "phycoerythrin"

≡ Red dinoflagellates (Ex. Gonyaulax) — divide very fast by binary
fission due to which their
number increases and sea
appears red (RED TIDES)

≡ Red dinoflagellates — Secrete toxins (eg. Saxitoxin) which kills other marine animals eg. fishes.

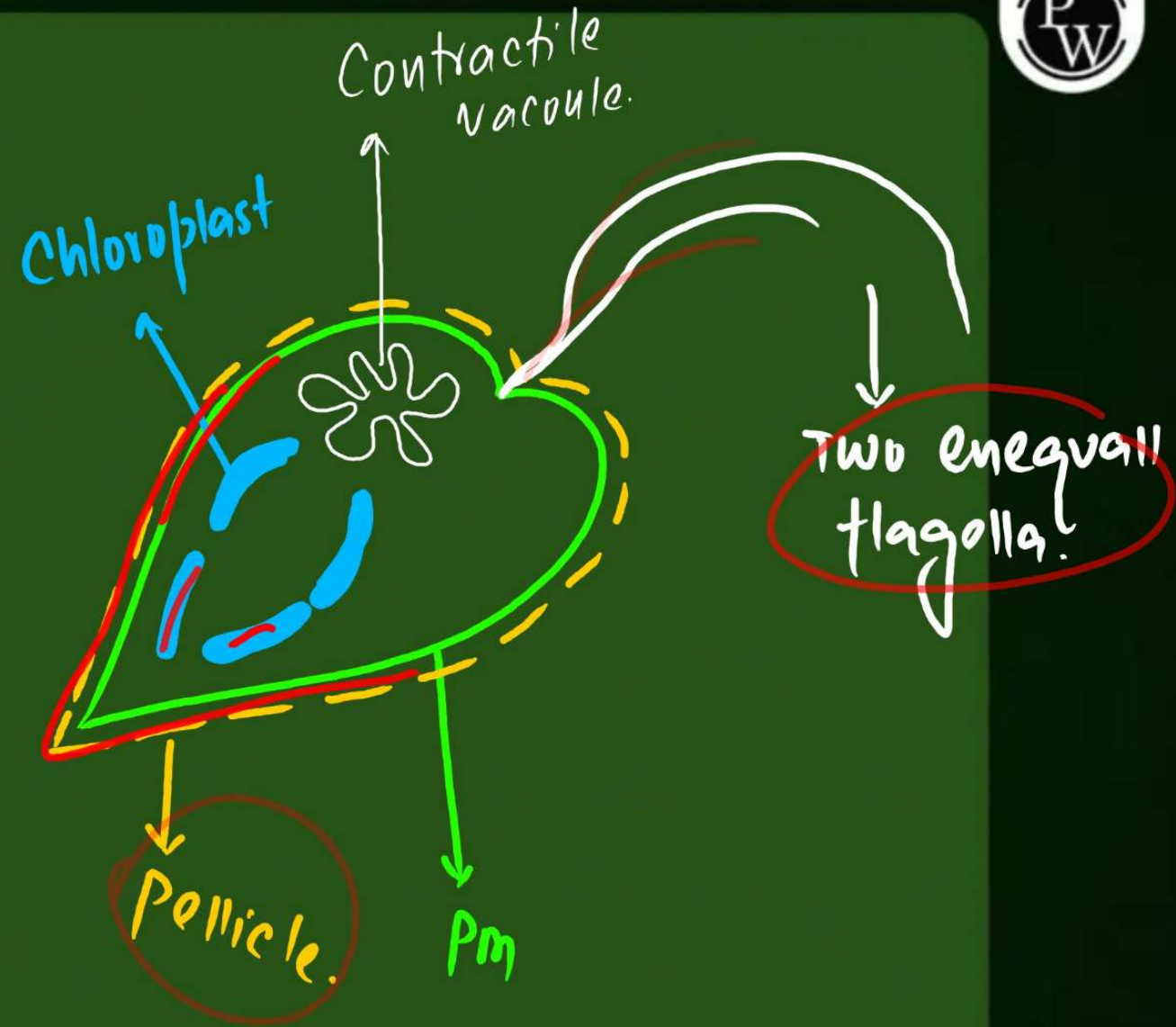
≡ Photosynthetic. ✓

≡ Pigments — Chla, Chlc, Xanthophyll.

EUGLENOIDS

Ex. Euglena

- Habitat: Fresh H_2O ($\downarrow \downarrow$ salt conc.)
Stagnant H_2O (جھپٹا پانی)
- Cell wall: - nt.
- outside pm is +nt flexible protein covering kld as pellicle.
↓
It makes Euglena flexible.



- Flagella : +nt.
 - Two in no.
 - Unequal.
 - Apical in position

• Photosynthetic pigments : Chla, chl b, Xanthophyll

Note :- presence of chl b is higher plant like character.

2. Euglena — Connecting link b/w plants and animals

No cell wall
(Animal like character)

presence of chl b
(higher plant like character).

Mode of Nutrition :

Mixotrophic

"In absence of light"

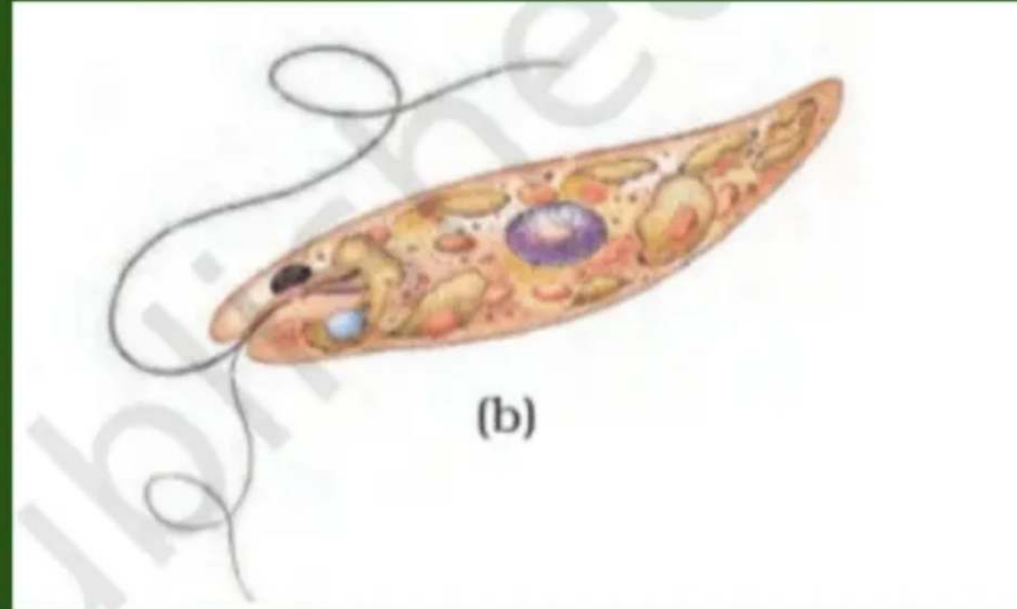
↓
Euglenoids act as
predators on small
organisms (Heterotrophic
Nutrition)

"In presence of light"

↓
they perform photosynthesis
(Autotrophic Nutrition)

• Contractile vacuule: helps in osmoregulation

↓
maintaining H_2O and ion concentration
inside body of organism.



Euglena

QUESTIONS

HW

In which group of organisms the cell walls form two thin overlapping shells which fit together?
[MR* 2015 Re]

- 1 Euglenoids
- 2 Dinoflagellates
- 3 Slime moulds
- 4 Chrysophytes

QUESTIONS

H/W

Select the wrong statement:

[2016 II]

- 1** Diatoms are chief producers in the oceans
- 2** Diatoms are microscopic and float passively in water
- 3** The walls of diatoms are easily destructible
- 4** 'Diatomaceous earth' is formed by the cell wall of diatoms.

QUESTIONS

H/w

Chrysophytes, Euglenoids, Dinoflagellates and Slime moulds are included in the kingdom: [2016 I]

- 1** Animalia
- 2** Monera
- 3** Protista
- 4** Fungi

H/w

QUESTIONS

Match List-I with List-II:

Choose the correct answer from the options given below:

[2024 Re]

1 A-III, B-IV, C-II, D-I

2 A-IV, B-II, C-I, D-III

3 A-IV, B-III, C-II, D-I

4 A-IV, B-II, C-III, D-I

List-I		List-II	
A.	Euglenoid	(I)	Parasitic
B.	Dinoflagellate	(II)	Saprophytic
C.	Slime mould	(III)	Photosynthetic
D.	<i>Plasmodium</i>	(IV)	Switching between photosynthetic and heterotrophic mode

QUESTIONS

H/W

Which of the following organisms possesses characteristics of a plant and an animal? [1995]

- 1 *Euglena*
- 2 *Paramecium*
- 3 Bacteria
- 4 *Mycoplasma*

Thank
You